

Predicting Personality from Text: A Dual-Task Psycholinguistic Analysis of Machine Learning and Transformer Models

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Abstract—Automatic Personality Prediction (APP) from text is an upcoming task in Natural Language Processing (NLP). It has a broad set of applications such as in personalized systems, social media analysis and empathetic dialogue agents. Our paper presents a rigorous comparative analysis of classical machine learning models (Random Forest, XGBoost, SVM) and a transformer based architecture for Myers-Briggs Type Indicator (MBTI) prediction using the MBTIBench dataset. A key contribution of our work is a dual task psycholinguistic evaluation framework which assesses models on both multi label classification of discrete MBTI types and multi output regression of continuous dimension scores. We also move beyond predictive accuracy and into explainability by employing SHAP to provide model interpretability. This analysis enables a psycholinguistic comparison of the top features driving predictions, validating how models map linguistic cues to psychological constructs. Experimental results show that traditional ensemble models such as XGBoost and Random Forest outperform transformer-based architectures in classification, achieving an F1 score of 0.589 and 0.550 respectively. Transformer models such as RoBERTa exhibit competitive results in regression with average mean squared errors around 0.088-0.089. The observed performance gap is mainly attributed to the limited size of the dataset which restricts the data hungry transformer models from fully leveraging their contextual understanding capabilities. We highlight the trade-offs between model complexity and data availability; establishing an interpretable baseline for future work on scalable, personality-aware NLP systems.

Index Terms—Automatic Personality Prediction (APP), Myers-Briggs Type Indicator (MBTI), transformers, ensemble learning, interpretability, explainability.

I. INTRODUCTION

Automatic personality prediction (APP) refers to the computational task of detecting an individual's personality traits based on their generated text [1]. It has become an area of active research in Natural Language Processing (NLP) with significant applications in personalized recommendations, social media analysis and empathetic dialogue systems. The task generally considers different personality frameworks and can be framed as classification or regression depending on whether discrete types or continuous trait scores are predicted. Myers-Briggs Type Indicator (MBTI) is particularly popular personality framework used for APP due to its structured four-dimensional categorization as shown in figure 1. By combining these four dichotomous dimensions, individuals can be classified into any one of sixteen distinct personality types such as INFP, ESTJ, ENFJ, etc.

Despite advances in NLP, the complexity of mapping linguistic cues to psychological theories remains a significant obstacle. Traditional machine learning approaches depend on fixed vector representations such as word embeddings which capture semantic similarity but overlook deeper contextual dependencies. In contrast, recent transformer models like BERT excel at understanding contextual relationships but require substantial computational resources and need to be carefully fine-tuned. Thus, this task is particularly relevant given the diverse linguistic expressions of personality features which

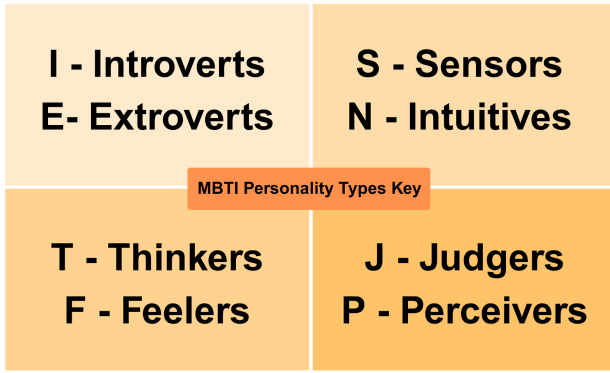


Fig. 1: Myers-Briggs personality types dimensions.

demand sophisticated modeling techniques capable of capturing semantic nuances and contextual information. Moreover, a considerable amount of prior research has utilized the popular MBTI dataset from Kaggle [2]. The primary issue with this is a fundamental sampling bias, as many users who contributed text to the dataset were likely already aware of their own MBTI type [3]. This self selection bias and potential for overfitting threaten the generalizability of practically all prior work, coming across a phenomenon called *Dataset Schism* [4].

Our study confronts this methodological challenge. We present our work not as another comparative analysis, but as a necessary and deliberate step to advance the reliability of APP. To do this, we leverage the MBTIBench dataset [2] to develop computational models for personality prediction. We do this by two complementary tasks: 1) classification of discrete MBTI types and 2) regression of continuous personality dimension scores. By grounding our experiments in this more robust dataset, we aim to establish a more trustworthy baseline for the field. We first pre-process the dataset to create a textual corpus which serve as input features for ensemble-based machine learning models. We compare classical machine learning models (XGBoost, Random Forest, SVM) with transformer models (BERT variants) to evaluate representation strategies for MBTI personality detection. The dual focus on both classification and regression tasks in our study enables a more comprehensive modeling of personality traits. Classification provides discrete personality labels which are useful for categorical analysis, while regression captures the continuum and intensity of traits, allowing for finer-grained interpretation.

Furthermore, building accurate models is only the first step. To truly advance APP and its real-world applications, we must also understand why models make their predictions. We go beyond performance metrics and employ SHAP (SHapley Additive exPlanations) [5] to provide robust interpretability. Critically, our contribution is not just to analyze feature importance but also incorporate explainability by conducting a psycholinguistic comparison of the top features that drive the model’s prediction for each of the four MBTI personality dimensions, as determined by SHAP analysis. This XAI-driven approach allows us to validate how models are learning and to what extent their internal logic aligns with established psycho-

logical theories of personality expression, providing a more transparent foundation for building trustworthy, empathetic systems.

II. RELATED WORK

Various studies have explored machine learning techniques to classify MBTI personality types from social media texts and other user-generated content [6]–[10]. Classification of MBTI types was performed using various pre-processing steps and traditional machine learning models on the ‘MBTI’ dataset on Kaggle. Another 2025 work proposed a machine learning framework leveraging Support Vector Machines (SVM) and Random Forest classifiers, achieving up to 77% accuracy by incorporating feature engineering methods such as sentiment analysis and lexical statistics [11]. Complementing these classical models, transformer-based approaches utilizing fine-tuned BERT and RoBERTa models have been investigated [12], which use explainable AI to reason how models process language features relevant to personality, uncovering theory-coherent patterns aligned with MBTI theory [3].

III. PROPOSED METHODOLOGY

A. Problem Formulation

The MBTIBench dataset [2] provides both soft labels that are probabilistic scores reflecting the likelihood of each MBTI dimension class and hard labels which are discrete classes for each dimension. The dataset covers all four MBTI dichotomies: Extraversion/Introversion (E-I), Sensing/Intuition (S-N), Thinking/Feeling (T-F), and Judging/Perceiving (J-P). This dual-label structure enables evaluation under both probabilistic and categorical prediction settings. Based on this, we define two complementary problems:

- 1) **Predictive Modeling:** This task is framed as a combination of multi-label classification and regression problem across the four MBTI dimensions.
 - For hard-labels, the objective is to predict a discrete class for each dimension.
 - For soft-labels, models are optimized to regress the underlying probability of class dimensions.
- 2) **Model Interpretability and Explainability:** This task is to interpret the trained models to understand the linguistic basis for their predictions. The objectives are to:
 - Determine which linguistic patterns or features are most influential in shaping model predictions for each of the MBTI dimensions.
 - Assess the relative importance or contribution of these features in guiding predictions toward one trait pole versus another.
 - Facilitate comparison of model-learned linguistic tendencies with known psycholinguistic or behavioral theories.

B. Models

We experiment using multiple machine learning ensembles and transformer-based models. Since our task requires to perform binary classification and regression across 4 different dimensions, we utilize Multi-output Classifier and Multi-output Regressor for the classification and regression tasks in ML models respectively.

1) *Multi-output Classifier*: Let the input feature embedding be $x \in \mathbb{R}^d$ and the MBTI labels be $y = \{y^{(1)}, y^{(2)}, y^{(3)}, y^{(4)}\}$, as per to the four MBTI dimensions.

Each dimension $k \in \{1, 2, 3, 4\}$ is treated as a binary classification task:

$$\hat{y}^{(k)} = f^{(k)}(x; \theta^{(k)}), \quad \hat{y}^{(k)} \in \{0, 1\}. \quad (1)$$

The multi-output classifier is defined as:

$$\hat{y} = (\hat{y}^{(1)}, \hat{y}^{(2)}, \hat{y}^{(3)}, \hat{y}^{(4)}). \quad (2)$$

The overall classification loss is the average of per-dimension binary cross-entropy losses:

$$\mathcal{L}_{\text{clf}} = \frac{1}{4} \sum_{k=1}^4 \ell_{\text{BCE}}(y^{(k)}, \hat{y}^{(k)}). \quad (3)$$

TABLE I: Per-Dimension F1 Scores for Classification Models

Model	E/I	S/N	T/F	J/P	Average
XGBoost	0.583	0.528	0.670	0.531	0.578
RandomForest	0.517	0.583	0.668	0.563	0.583
SVM	0.341	0.464	0.333	0.353	0.373
BERT	0.510	0.460	0.640	0.540	0.538
RoBERTa	0.360	0.490	0.680	0.480	0.503
DeBERTa	0.370	0.420	0.670	0.520	0.495

TABLE II: Per-Dimension MSE for Regression Models

Model	E/I	S/N	T/F	J/P	Average
XGBoost	0.072	0.124	0.096	0.109	0.100
RandomForest	0.084	0.110	0.080	0.094	0.092
SVM	0.089	0.112	0.080	0.096	0.094
BERT	0.090	0.098	0.081	0.089	0.089
RoBERTa	0.086	0.095	0.075	0.099	0.089
DeBERTa	0.092	0.094	0.078	0.102	0.091

2) *Multi-output Regressor*: For the regression setting, each dimension is represented as a soft label $p^{(k)} \in [0, 1]$, denoting the probability distribution of class membership.

The predicted value for dimension k is:

$$\hat{p}^{(k)} = g^{(k)}(x; \theta^{(k)}), \quad \hat{p}^{(k)} \in [0, 1]. \quad (4)$$

The multi-output regressor is then:

$$\hat{p} = (\hat{p}^{(1)}, \hat{p}^{(2)}, \hat{p}^{(3)}, \hat{p}^{(4)}). \quad (5)$$

The regression loss can be defined as Mean Squared Error (MSE):

$$\mathcal{L}_{\text{reg}} = \frac{1}{4} \sum_{k=1}^4 (p^{(k)} - \hat{p}^{(k)})^2, \quad (6)$$

For the transformer-based models, we adopt a dual-task architecture which consists of a shared encoder and two task-specific decoders for regression and classification.

An input sequence x is encoded using a pretrained transformer, producing the [CLS] representation $h \in \mathbb{R}^d$:

$$h = \text{Transformer}(x)_{[\text{CLS}]}. \quad (7)$$

Two parallel feed-forward heads are applied:

$$\hat{p} = W_r \sigma(W_r' h + b_r') + b_r, \quad \hat{p} \in \mathbb{R}^4, \quad (8)$$

$$\hat{y} = W_c \sigma(W_c' h + b_c') + b_c, \quad \hat{y} \in \mathbb{R}^4, \quad (9)$$

where $\hat{p}$ are regression outputs (soft labels) and $\hat{y}$ are classification logits (hard labels).

IV. EXPERIMENTATION

A. Dataset

The experiments were conducted using the MBTIBench dataset [2], which contains user-generated posts annotated with both soft labels and hard labels. Each record includes a user identifier, data source and a collection of text posts concatenated to form a single textual input. Train/test split was 80 : 20.

B. Preprocessing

The data were preprocessed with transformations of lowercasing, removal of special characters and normalization of whitespace. The textual data were tokenized into word sequences and used for generating dense vector representations through TF-IDF vectorized embeddings. Stopword removal was intentionally avoided to retain stylistic and lexical cues indicative of personality traits.

C. Word Embeddings for Machine Learning Models

To construct interpretable and linguistically rich text representations, we used a Term Frequency–Inverse Document Frequency (TF-IDF) based feature extraction pipeline. This approach transforms each user’s concatenated text post into a weighted numerical vector that captures the relative importance of words and phrases across the corpus.

We utilized unigrams, bigrams and trigrams to encode both individual words and short psycholinguistic expressions such as “I feel that” or “it seems like” which often convey personality-related cues. Sublinear scaling of term frequencies was used to lessen the impact of high frequency tokens. The feature space was restricted to 5000 dimensions to maintain computational efficiency while preserving sufficient linguistic diversity.

The resulting TF-IDF matrix had a shape of (286, 5000) since the dataset contains 286 text samples that are represented by 5000 feature dimensions.

D. Model Parameters

1) Traditional Machine Learning Models:

- **Random Forest:** Configured with 400 trees and a maximum depth of 12 with a minimum of 8 samples for node splitting and 4 samples per leaf. Feature selection follows the $\sqrt{n_features}$ rule with class weighting adjusted for balance.
- **XGBoost:** Utilized 250 boosting rounds with a tree depth of 4, a learning rate of 0.08 and subsampling/column sampling ratios of 0.85. Regularization terms were set as $\alpha = 0.2$ and $\lambda = 1.5$.
- **Support Vector Machine (SVM):** Implemented using the RBF kernel with $C = 15$, $\gamma = \text{scale}$ and $\epsilon = 0.05$.

2) Transformer-based Architectures:

- **Base Models:** Experiments were conducted with bert-base [13], roberta-base [14], and deberta-v3-base [15].
- **Training Setup:** The maximum input sequence length was fixed at 256 with a batch size of 16 and training carried out for 7 epochs.
- **Optimization Details:** The AdamW optimizer was used with a learning rate of 5×10^{-5} and dropout probability of 0.3.
- **Composite Loss:** The training objective combined mean squared error and binary cross-entropy losses, expressed as:

$$\mathcal{L} = \alpha \cdot \text{MSE} + (1 - \alpha) \cdot \text{BCE}$$

V. RESULTS

Tables I and II summarize the experimental results for the classification and regression tasks respectively. Performance across the four MBTI dimensions was evaluated to assess each model’s consistency and robustness.

A. Classification

According to Table I, traditional ML models achieved competitive performance compared to transformer models. Among all classification models, XGBoost marked the highest F1 metric (0.589) succeeded by Random Forest (0.550) and SVM (0.563). The T/F dimension consistently yielded the highest F1 scores across models. This suggested that textual cues distinguishing logical versus empathetic reasoning styles are the most separable. Conversely, the S/N and J/P dimensions exhibited relatively lower scores, indicating higher overlap in linguistic features between the paired traits. In contrast, transformer models such as BERT, RoBERTa and DeBERTa achieved lower average F1 scores (0.45–0.54). This decrease is primarily caused by the limited dataset size which restricts the models’ ability to leverage their large parameter spaces effectively. Transformers typically require larger and more diverse datasets to generalize well while ensemble models are perform better under data-scarce conditions.

B. Regression

Table II summarizes the Mean Squared Error (MSE) results for soft personality prediction. Random Forest attained the lowest average MSE (0.087) closely followed by RoBERTa (0.089). These results indicate that both traditional ML and transformer models can calculate personality scores on a continuous scale with reasonable accuracy. Consistent with the classification results, the T/F dimension showed the lowest MSE values across all models while S/N and J/P displayed higher errors. This reflected more subtle lexical differences between these paired dimensions.

VI. INTERPRETABILITY AND PSYCHOLINGUISTIC ANALYSIS

To understand the linguistic signals guiding personality predictions, we developed a hybrid interpretability pipeline combining SHAP feature attribution [5] with Empath psycholinguistic categorization [16]. The objective was to look beyond the raw model weights and identify lexical and semantic cues that distinguish the poles of each MBTI dimension.

We computed SHAP [5] values for the best performing model (XGBoost) corresponding to the MBTI dimensions. For each dimension, the top 500 n -grams with the highest mean absolute SHAP magnitudes were extracted separately for the poles (positive and negative). This provided an interpretable ranking of predictive linguistic patterns. Each n -gram was then decomposed into its constituent tokens with SHAP weights propagated to the individual word level to preserve contribution proportionality.

To ground these lexical features in interpretable psychological semantics, we employed the Empath lexicon [16] which maps words into 200 affective and topical categories (e.g., *social*, *work*, *family*, *optimism*). For each MBTI pole, we constructed Empath profile by summing category frequencies weighted by their SHAP-weighted importance and normalizing to yield proportionate psycholinguistic distributions.

TABLE III: Top 5 Psycholinguistic Categories by Empath Profiles for each MBTI Dimension

Dim.	Category	Pos. Pole	Neg. Pole	χ^2	p-value	Dominant
E/I	envy	0.0119	0.0040	0.0040	0.9497	E
E/I	hate	0.0181	0.0101	0.0023	0.9615	E
E/I	death	0.0164	0.0101	0.0015	0.9691	E
E/I	giving	0.0766	0.0890	0.0010	0.9746	I
E/I	speaking	0.0578	0.0663	0.0006	0.9801	I
S/N	writing	0.0122	0.0207	0.0022	0.9627	N
S/N	children	0.0139	0.0081	0.0015	0.9687	S
S/N	childish	0.0255	0.0186	0.0011	0.9734	S
S/N	communication	0.0394	0.0460	0.0005	0.9816	N
S/N	speaking	0.0324	0.0374	0.0004	0.9847	N
T/F	swearing terms	0.0190	0.0099	0.0029	0.9573	T
T/F	trust	0.0518	0.0434	0.0008	0.9777	T
T/F	achievement	0.0145	0.0192	0.0006	0.9797	F
T/F	social media	0.0303	0.0248	0.0006	0.9812	T
T/F	help	0.0080	0.0107	0.0004	0.9838	F
J/P	messaging	0.0205	0.0053	0.0092	0.9238	J
J/P	computer	0.0151	0.0036	0.0072	0.9324	J
J/P	family	0.0181	0.0286	0.0024	0.9606	P
J/P	communication	0.0530	0.0396	0.0020	0.9640	J
J/P	healing	0.0139	0.0221	0.0019	0.9650	P

The results in Table III reveal distinctive psycholinguistic patterns across MBTI dimensions. For the *E/I* (Extraver-

S/N word cloud



sion/Introversion) dimension the extraversion pole's associations with categories such as *envy*, *hate* and *death* were driven by words like "hate", "frustration" and "life", highlighting emotionally expressive language patterns. Conversely, the introversion pole's elevated categories including *giving* and *speaking*, were grounded in words like "thank", "guide", "explain" and "word", reflecting socially restrained tendencies.

Similarly, for the *S/N* (Sensing-Intuition) axis, intuition-oriented profiles emphasized categories like *messaging*, *computer* and *communication* with key words “type”, “email” and “answer” consistent with abstract or mediated forms of interaction. Sensing-oriented profiles were instead characterized by categories such as *family* and *healing* represented by words like “family”, “better” and “forgive”, indicating concrete and experience-driven discourse.

For *T/F* (Thinking-Feeling), feeling-oriented users showed prominence in affective categories (*children, childish*) with words such as “happy”, “fun” and “little”, while thinking-oriented poles emphasized analytic and informational categories (*communication, speaking*) with terms like “explain”, “word” and “idea”.

Finally, for the *J/P* (Judging-Perceiving) dimension, perceiving-oriented profiles highlighted emotionally intense categories including *hate*, *envy* and *death*, supported by words like “hate”, “life” and “frustration.” Judging-oriented profiles instead exhibited structured or socially regulated categories exemplified by *giving* and *communication*, with representative words “thank”, “order” and “agree”. This analysis illustrates how SHAP feature attribution and Empath profiling allows decomposition of abstract personality traits into interpretable lexical signals and thus, grounding statistical patterns in actual word usage.

Finally, we performed a chi-square test of independence

between positive and negative pole profiles across categories to quantify which linguistic themes significantly differentiate opposing traits. Across all dimensions, the χ^2 values were relatively low and the corresponding p-values were consistently high (all > 0.92). This indicates that the observed differences between positive and negative pole profiles are not statistically significant at conventional thresholds. While this suggests that the individual category distributions alone do not strongly discriminate between opposing traits, the directionality of the differences towards the dominant dimension still reveals meaningful trends. For instance, the extraversion pole (*E*) shows higher relative weights in emotionally expressive categories like *envy*, *hate* and *death*, whereas the introversion pole (*I*) is more associated with interpersonal or reflective categories such as *giving* and *speaking*. Similar directional patterns are evident for the other MBTI dimensions, highlighting interpretable psycholinguistic tendencies despite the lack of strong statistical significance at the level of individual categories.

VII. CONCLUSION & FUTURE WORK

This paper presented a rigorous comparative analysis of classical machine learning ensembles and transformer-based models for the task of Automatic Personality Prediction (APP) from text, specifically focusing on the Myers-Briggs Type Indicator (MBTI). The work was motivated by a critical need to address methodological shortcomings of over-reliance on biased datasets and evaluation frameworks that fail to capture the nuanced, continuous nature of psychological traits. By leveraging a next-generation dataset such as MBTIBench [2] and a dual-task evaluation framework, our work provides a clear-eyed assessment of the current capabilities and limitations of both classical and transformer models. This helped

establish a more reliable and psychologically valid benchmark for future investigations.

Our experiments demonstrated that ensemble models, particularly XGBoost, outperform transformer based architectures in classification in low-resource settings. However, transformers like RoBERTa remain competitive for the regression task. This suggests that they are promising for capturing the nuanced nature of personality traits. Our key contributions include explainability focused psycholinguistic analysis using SHAP and Empath, which provided interpretable insights into the specific lexical cues models associate with each personality dimension. This establishes a more reliable and psychologically valid benchmark for future investigations. While transformer models seem promising for capturing the fine-grained nuances of personality encoded in language, their full potential can only be unlocked through the development of larger datasets.

ACKNOWLEDGMENT

We acknowledge the MBTIBench research team for making the dataset publicly available which played a crucial role in enabling this study. Finally, we extend our appreciation to the faculty mentors and peers who provided valuable feedback during the experimentation and writing process.

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